

Grid Maps and Mapping with Known Poses

Autonomous Navigation of Mobile Robots

Anna Semakova

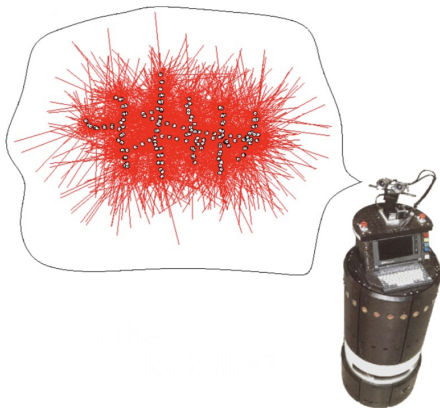
`a.semakova@gmail.com`

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Why Mapping?

- Learning maps is one of the fundamental problems in mobile robotics
- Maps allow robots to efficiently carry out their tasks, allow localization ...
- Successful robot systems rely on maps for localization, path planning, activity planning etc.

The General Problem of Mapping



What does the environment look like?

The General Problem of Mapping

- Formally, mapping involves, given the sensor data

$$d = \{u_1, z_1, u_2, z_2, \dots, u_t, z_t\}$$

- to calculate the most likely map

$$m^* = \operatorname{argmax}_m P(m \mid d)$$

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- We'll describe [how to calculate a map given the robot's poses](#)

The General Problem of Mapping with Known Poses

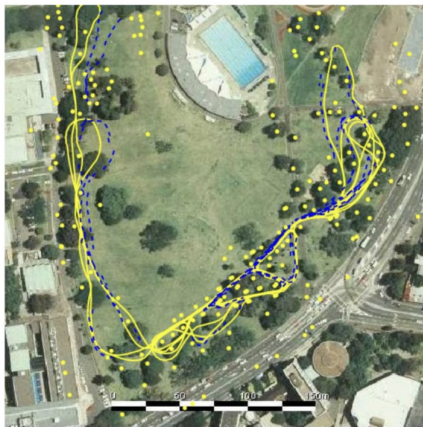
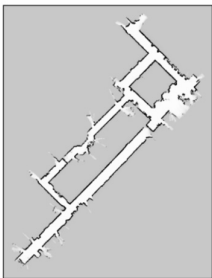
- Formally, mapping with known poses involves, given the measurements and the poses

$$d = \{x_1, z_1, x_2, z_2, \dots, x_t, z_t\}$$

- to calculate the most likely map

$$m^* = \operatorname{argmax}_m P(m \mid d)$$

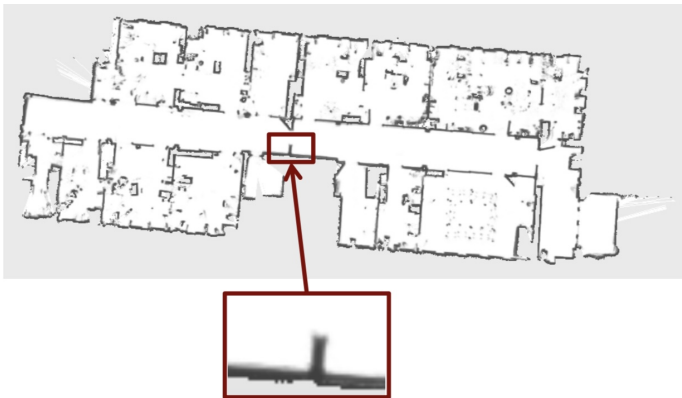
Features vs. Volumetric Maps



Grid Maps

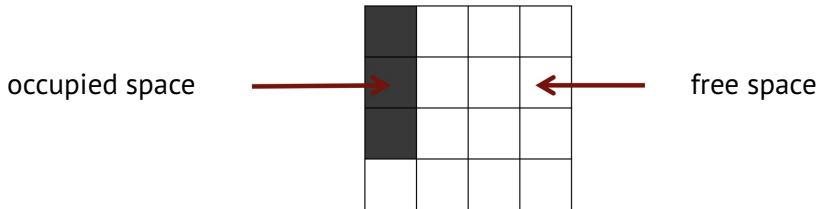
- We discretize the world into cells
- The grid structure is rigid
- Each cell is assumed to be occupied or free
- It is a non-parametric model
- It requires substantial memory resources
- It does not rely on a feature detector

Example



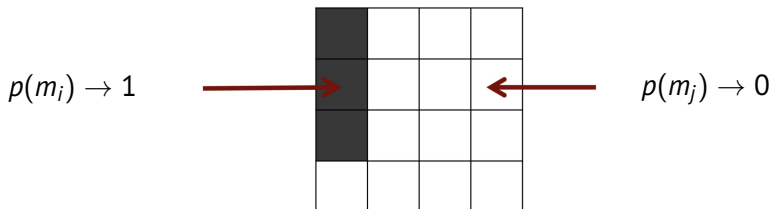
Assumption 1

- The area that corresponds to a cell is either completely free or occupied



Representation

- Each cell is a **binary random variable** that models the occupancy



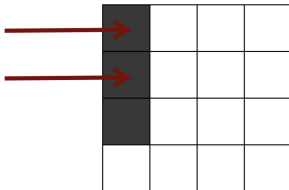
Occupancy Probability

- Each cell is a **binary random variable** that models the occupancy
- Cell is occupied $p(m_i) = 1$
- Cell is not occupied $p(m_i) = 0$
- No information $p(m_i) = 0.5$
- The environment is assumed to be **static**

Assumption 2

- The cells (the random variables) are **independent** of each other

no dependency
between the cells



Representation

- The probability distribution of the map is given by the product of the probability distributions of the individual cells

$$p(m) = \prod_i p(m_i)$$

↑ ↑

map cell

Representation

- The probability distribution of the map is given by the product of the probability distributions of the individual cells

$$p(m) = \prod_i p(m_i)$$



four-dimensional vector



four independent cells

Estimating a Map From Data

- Given sensor data $z_{1:t}$ and the poses $x_{1:t}$ of the sensor, estimate the map

$$p(m | z_{1:t}, x_{1:t}) = \prod_i p(m_i | z_{1:t}, x_{1:t})$$

↑

binary random variable

⇒ Binary Bayes filter (for a static state)

Static State Binary Bayes Filter

$$p(m_i | z_{1:t}, x_{1:t}) \stackrel{\text{Bayes rule}}{=} \frac{p(z_t | m_i, z_{1:t-1}, x_{1:t}) p(m_i | z_{1:t-1}, x_{1:t})}{p(z_t | z_{1:t-1}, x_{1:t})}$$

Static State Binary Bayes Filter

$$\begin{aligned} p(m_i | z_{1:t}, x_{1:t}) &\stackrel{\text{Bayes rule}}{=} \frac{p(z_t | m_i, z_{1:t-1}, x_{1:t}) p(m_i | z_{1:t-1}, x_{1:t})}{p(z_t | z_{1:t-1}, x_{1:t})} \\ &\stackrel{\text{Markov}}{=} \frac{p(z_t | m_i, x_t) p(m_i | z_{1:t-1}, x_{1:t-1})}{p(z_t | z_{1:t-1}, x_{1:t})} \end{aligned}$$

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Static State Binary Bayes Filter

$$\begin{aligned} p(m_i | z_{1:t}, x_{1:t}) &\stackrel{\text{Bayes rule}}{=} \frac{p(z_t | m_i, z_{1:t-1}, x_{1:t}) p(m_i | z_{1:t-1}, x_{1:t})}{p(z_t | z_{1:t-1}, x_{1:t})} \\ &\stackrel{\text{Markov}}{=} \frac{p(z_t | m_i, x_{1:t}) p(m_i | z_{1:t-1}, x_{1:t-1})}{p(z_t | z_{1:t-1}, x_{1:t})} \\ &\stackrel{\text{Bayes rule}}{=} \frac{p(m_i | z_t, x_t) p(z_t | x_t) p(m_i | z_{1:t-1}, x_{1:t-1})}{p(m_i | x_t) p(z_t | z_{1:t-1}, x_{1:t})} \end{aligned}$$

Static State Binary Bayes Filter

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Static State Binary Bayes Filter

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Do exactly the same for the opposite event

$$p(\neg m_i | z_{1:t}, x_{1:t}) \stackrel{\text{The same}}{=} \frac{p(\neg m_i | z_t, x_t) p(z_t | x_t) p(\neg m_i | z_{1:t-1}, x_{1:t-1})}{p(\neg m_i) p(z_t | z_{1:t-1}, x_{1:t})}$$

Static State Binary Bayes Filter

- By computing the ratio of both probabilities, we obtain

$$\frac{p(m_i | z_{1:t}, x_{1:t})}{p(\neg m_i | z_{1:t}, x_{1:t})} = \frac{\frac{p(m_i | z_t, x_t) \cancel{p(z_t | x_t)} p(m_i | z_{1:t-1}, x_{1:t-1})}{p(m_i) \cancel{p(z_t | z_{1:t-1}, x_{1:t})}}}{\frac{p(\neg m_i | z_t, x_t) \cancel{p(z_t | x_t)} p(\neg m_i | z_{1:t-1}, x_{1:t-1})}{p(\neg m_i) \cancel{p(z_t | z_{1:t-1}, x_{1:t})}}}$$

Static State Binary Bayes Filter

- By computing the ratio of both probabilities, we obtain

$$\begin{aligned} & \frac{p(m_i | z_{1:t}, x_{1:t})}{p(\neg m_i | z_{1:t}, x_{1:t})} \\ &= \frac{p(m_i | z_t, x_t) p(m_i | z_{1:t-1}, x_{1:t-1}) p(\neg m_i)}{p(\neg m_i | z_t, x_t) p(\neg m_i | z_{1:t-1}, x_{1:t-1}) p(m_i)} \\ &= \frac{p(m_i | z_t, x_t)}{1 - p(m_i | z_t, x_t)} \frac{p(m_i | z_{1:t-1}, x_{1:t-1})}{1 - p(m_i | z_{1:t-1}, x_{1:t-1})} \frac{1 - p(m_i)}{p(m_i)} \end{aligned}$$

Static State Binary Bayes Filter

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$$\begin{aligned} & \frac{p(m_i | z_{1:t}, x_{1:t})}{1 - p(m_i | z_{1:t}, x_{1:t})} \\ &= \frac{p(m_i | z_t, x_t) p(m_i | z_{1:t-1}, x_{1:t-1}) p(\neg m_i)}{p(\neg m_i | z_t, x_t) p(\neg m_i | z_{1:t-1}, x_{1:t-1}) p(m_i)} \\ &= \underbrace{\frac{p(m_i | z_t, x_t)}{1 - p(m_i | z_t, x_t)}}_{\text{uses } z_t} \underbrace{\frac{p(m_i | z_{1:t-1}, x_{1:t-1})}{1 - p(m_i | z_{1:t-1}, x_{1:t-1})}}_{\text{recursive term}} \underbrace{\frac{1 - p(m_i)}{p(m_i)}}_{\text{prior}} \end{aligned}$$

Occupancy Update Rule

■ Recursive rule

$$\frac{p(m_i | z_{1:t}, x_{1:t})}{1 - p(m_i | z_{1:t}, x_{1:t})} = \underbrace{\frac{p(m_i | z_t, x_t)}{1 - p(m_i | z_t, x_t)}}_{\text{uses } z_t} \underbrace{\frac{p(m_i | z_{1:t-1}, x_{1:t-1})}{1 - p(m_i | z_{1:t-1}, x_{1:t-1})}}_{\text{recursive term}} \underbrace{\frac{1 - p(m_i)}{p(m_i)}}_{\text{prior}}$$

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■ Often written as

$$Bel(m_t^i) = \left[1 + \frac{1 - p(m_t^i | z_t, x_t)}{p(m_t^i | z_t, x_t)} \frac{p(m_t^i)}{1 - p(m_t^i)} \frac{1 - Bel(m_{t-1}^i)}{Bel(m_{t-1}^i)} \right]^{-1}$$

Log Odds Notation

- Log odds ratio is defined as

$$l(x) = \log \frac{p(x)}{1 - p(x)}$$

- and with the ability to retrieve $p(x)$

$$p(x) = 1 - \frac{1}{1 + \exp l(x)}$$

Occupancy Mapping

in Log Odds Form

- The product turns into a sum

$$l(m_i | z_{1:t}, x_{1:t}) = l(m_i | z_t, x_t) + l(m_i | z_{1:t-1}, x_{1:t-1}) - l(m_i)$$

inv. sensor model recursive term prior

- or in short

$$l_{t,i} = \text{inv_sensor_model}(m_i, x_t, z_t) + l_{t-1,i} - l_0$$

Occupancy Mapping Algorithm

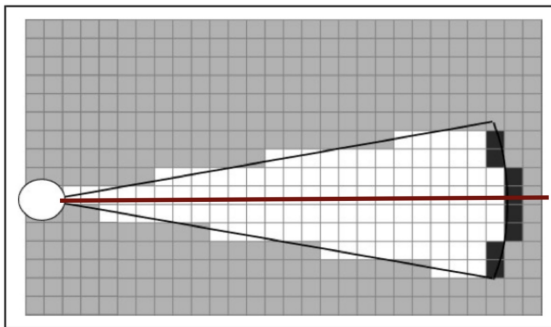
```
Algorithm occupancy_grid_mapping( $\{l_{t-1,i}\}, x_t, z_t$ ) :  
  For all cells  $m_i$  Do  
    If  $m_i$  in perceptual field of  $z_t$  Then  
       $l_{t,i} = l_{t-1,i} + \text{inv\_sensor\_model}(m_i, x_t, z_t) - l_0$   
    Else  
       $l_{t,i} = l_{t-1,i}$   
    Endif  
  Endfor  
Return  $\{l_{t,i}\}$ 
```

Highly efficient, only requires to compute sums

Occupancy Grid Mapping

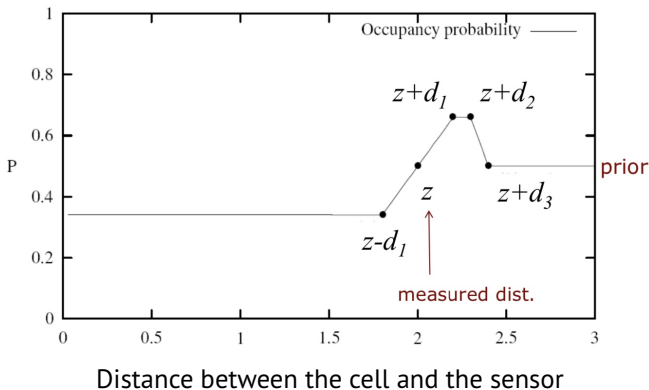
- Developed in the mid 80's by Moravec and Elfes
- Originally developed for noisy sonars
- Also called “mapping with know poses”

Inverse Sensor Model for Sonars Range Sensors

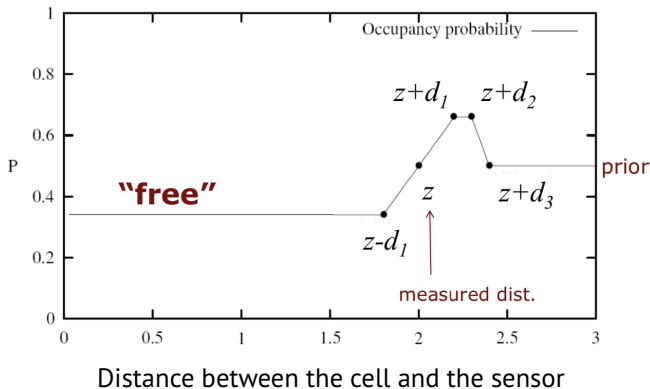


In the following, consider the cells along the optical axis (red line)

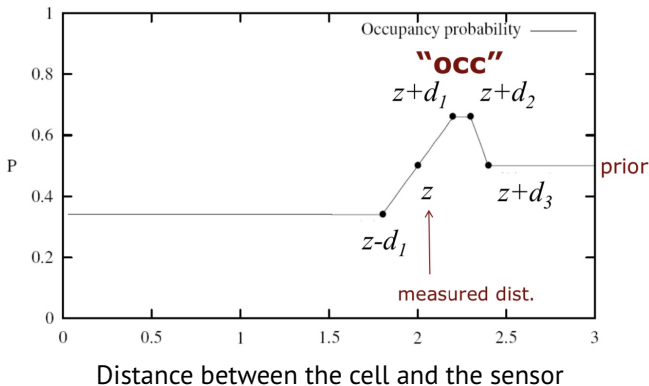
Occupancy Value Depending on the Measured Distance



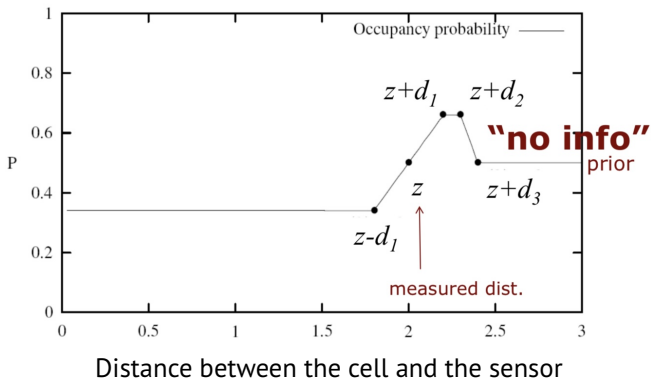
Occupancy Value Depending on the Measured Distance



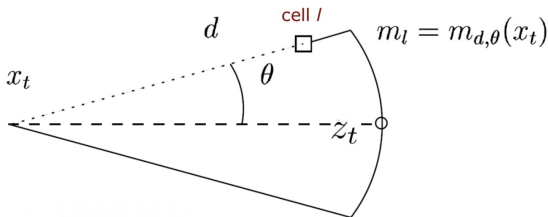
Occupancy Value Depending on the Measured Distance



Occupancy Value Depending on the Measured Distance

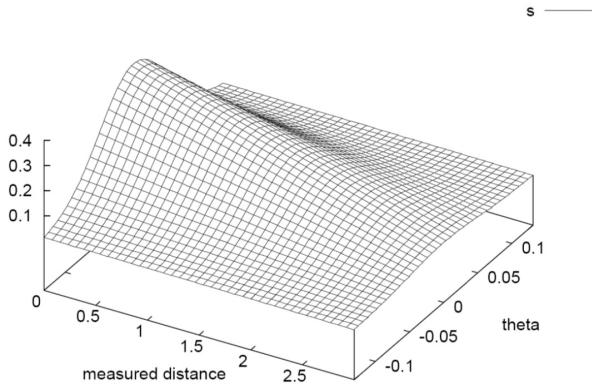


Update depends on the Measured Distance and Deviation from the Optical Axis



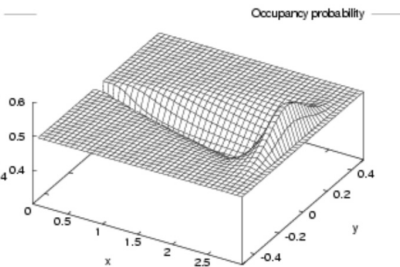
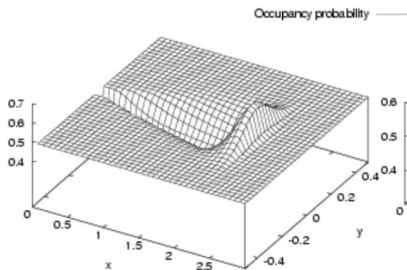
- Linear in z_t
- Gaussian in θ

Intensity of the Update



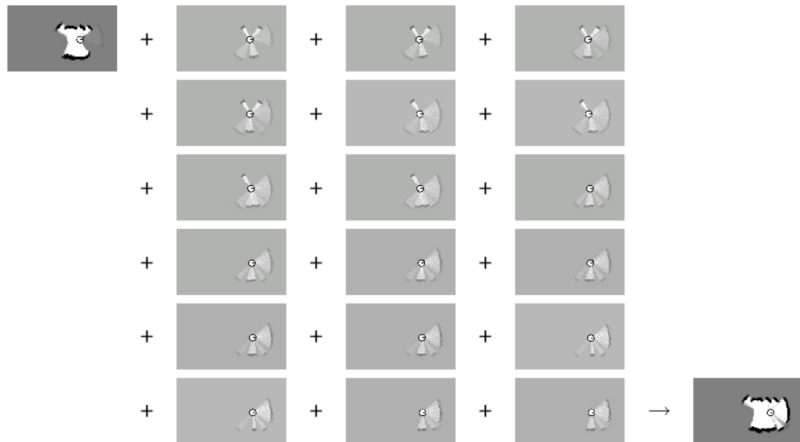
Resulting Model

$$p(m_i | z_t, x_t)$$



Example

Incremental Updating of Occupancy Grids

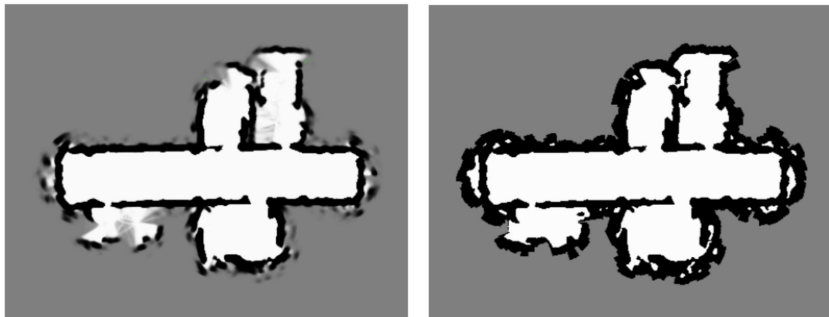


Resulting Map

Obtained with Ultrasound Sensors

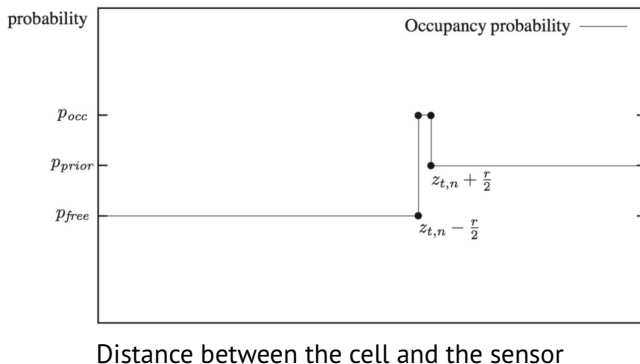


Resulting Occupancy and Maximum Likelihood Map



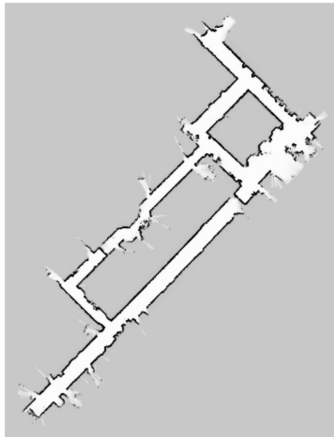
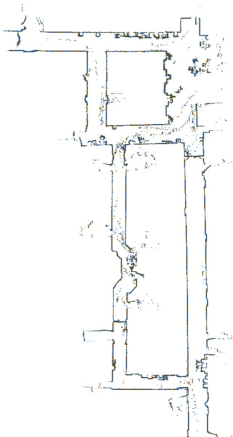
The maximum likelihood map is obtained by rounding the probability for each cell to 0 or 1

Inverse Sensor Model for Laser Range Finders



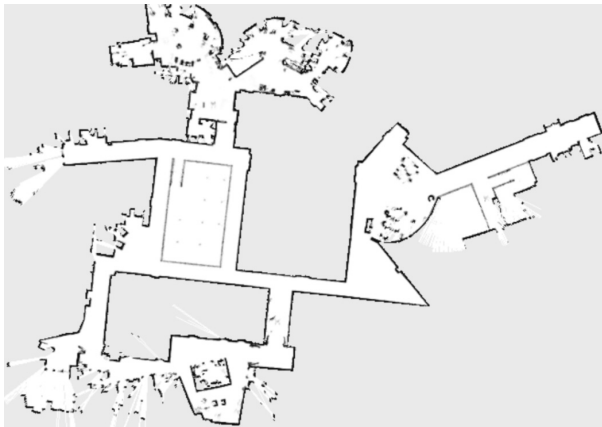
Occupancy Grids

from Laser Scans to Maps



Example

MIT CSAIL 3rd Floor



Alternative Counting Model

- For every cell count

$\text{hits}(x, y)$ – number of cases where a beam ended at $\langle x, y \rangle$

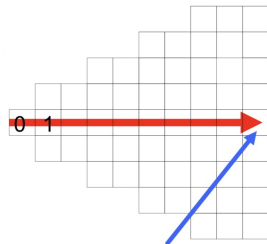
$\text{misses}(x, y)$ – number of cases where a beam passed through $\langle x, y \rangle$

$$\text{Bel}(m^{[xy]}) = \frac{\text{hits}(x, y)}{\text{hits}(x, y) + \text{misses}(x, y)}$$

- Value of interest – $P(\text{reflects}(x, y))$

The Measurement Model

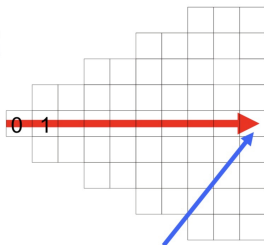
- Pose at time t – x_t
- Beam n of scan at time t – $z_{t,n}$
- Maximum range reading – $\zeta_{t,n} = 1$
- Beam reflected by an object – $\zeta_{t,n} = 0$



$m_f(x_t, n, z_{t,n})$
measured dist. in #cells

The Measurement Model

- Pose at time t – x_t
- Beam n of scan at time t – $z_{t,n}$
- Maximum range reading – $\zeta_{t,n} = 1$
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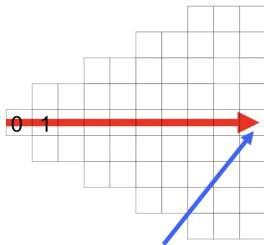
$$m_f(x_t, n, z_{t,n})$$

Max range – “first $z_{t,n} - 1$ cells covered by the beam must be free”

$$p(z_{t,n} | x_t, m) = \begin{cases} \prod_{k=0}^{z_{t,n}-1} (1 - m_f(x_t, n, k)) & \text{if } \zeta_{t,n} = 1 \\ 0 & \text{if } \zeta_{t,n} = 0 \end{cases}$$

The Measurement Model

- Pose at time t – x_t
- Beam n of scan at time t – $z_{t,n}$
- Maximum range reading – $\zeta_{t,n} = 1$
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$$m_f(x_t, n, z_{t,n})$$

Max range – “first $z_{t,n} - 1$ cells covered by the beam must be free”

$$p(z_{t,n} | x_t, m) = \begin{cases} \prod_{k=0}^{z_{t,n}-1} (1 - m_f(x_t, n, k)) & \text{if } \zeta_{t,n} = 1 \\ m_f(x_t, n, z_{t,n}) \prod_{k=0}^{z_{t,n}-1} (1 - m_f(x_t, n, k)) & \text{if } \zeta_{t,n} = 0 \end{cases}$$

Otherwise – “last cell reflected beam, all others free”

Computing the Most Likely Map

- Compute values for m that maximize

$$m^* = \operatorname{argmax}_m P(m \mid z_1, \dots, z_t, x_1, \dots, x_t)$$

- Assuming a uniform prior probability for $P(m)$, this is equivalent to maximizing

$$\begin{aligned} m^* &= \operatorname{argmax}_m P(z_1, \dots, z_t \mid m, x_1, \dots, x_t) \\ &= \operatorname{argmax}_m \prod_{t=1}^T P(z_t \mid m, x_t) \text{ since } z_t \text{ only depend on } x_t \\ &= \operatorname{argmax}_m \sum_{t=1}^T \ln P(z_t \mid m, x_t) \end{aligned}$$

Computing the Most Likely Map

cells beams

$$m^* = \operatorname{argmax}_m \sum_{j=1}^J \sum_{t=1}^T \sum_{n=1}^N (I(f(x_t, n, z_{t,n}) = j) \cdot (1 - \zeta_{t,n}) \cdot \ln m_j \\ + \sum_{k=0}^{z_{t,n}-1} I(f(x_t, n, k) = j) \cdot \ln(1 - m_j))$$

Computing the Most Likely Map

“beam n ends in cell j ”

$$m^* = \operatorname{argmax}_m \sum_{j=1}^J \sum_{t=1}^T \sum_{n=1}^N (I(f(x_t, n, z_{t,n}) = j) \cdot (1 - \zeta_{t,n}) \cdot \ln m_j \\ + \sum_{k=0}^{z_{t,n}-1} I(f(x_t, n, k) = j) \cdot \ln(1 - m_j))$$

Computing the Most Likely Map

“beam n ends in cell j ”

$$m^* = \operatorname{argmax}_m \sum_{j=1}^J \sum_{t=1}^T \sum_{n=1}^N (l(f(x_t, n, z_{t,n}) = j) \cdot (1 - \zeta_{t,n}) \cdot \ln m_j \\ + \sum_{k=0}^{z_{t,n}-1} l(f(x_t, n, k) = j) \cdot \ln(1 - m_j))$$

“beam n traversed cell j ”

Computing the Most Likely Map

“beam n ends in cell j ”

$$m^* = \operatorname{argmax}_m \sum_{j=1}^J \sum_{t=1}^T \sum_{n=1}^N (I(f(x_t, n, z_{t,n}) = j) \cdot (1 - \zeta_{t,n}) \cdot \ln m_j \\ + \sum_{k=0}^{z_{t,n}-1} I(f(x_t, n, k) = j) \cdot \ln(1 - m_j))$$

“beam n traversed cell j ”

Define

$$\alpha_j = \sum_{t=1}^T \sum_{n=1}^N I(f(x_t, n, z_{t,n}) = j) \cdot (1 - \zeta_{t,n})$$

$$\beta_j = \sum_{t=1}^T \sum_{n=1}^N \sum_{k=0}^{z_{t,n}-1} I(f(x_t, n, k) = j)$$

Meaning of α_j and β_j

- $\alpha_j = \sum_{t=1}^T \sum_{n=1}^N I(f(x_t, n, z_{t,n}) = j) \cdot (1 - \zeta_{t,n})$

Corresponds to the number of times a beam that is **not a maximum range beam ended in cell j** – hits(j)

- $\beta_j = \sum_{t=1}^T \sum_{n=1}^N \sum_{k=0}^{z_{t,n}-1} I(f(x_t, n, k) = j)$

Corresponds to the number of times a **beam traversed cell j without ending in it** – misses(j)

Computing the Most Likely Map

Accordingly, we get

$$m^* = \operatorname{argmax}_m \sum_{j=1}^J (\alpha_j \ln m_j + \beta_j \ln(1 - m_j))$$

As the m_j 's are independent we can maximize this sum by maximizing it for every j

If we set

we obtain

$$\frac{\partial}{\partial m_j} = \frac{\alpha_j}{m_j} - \frac{\beta_j}{1 - m_j} = 0 \quad m_j = \frac{\alpha_j}{\alpha_j + \beta_j}$$

Computing the most likely map reduces to counting how often a cell has reflected a measurement and how often the cell was traversed by a beam

Difference between Occupancy Grid Maps and Counting

- The counting model determines how often a cell reflects a beam
- The occupancy model represents whether or not a cell is occupied by an object
- Although a cell might be occupied by an object, the reflection probability of this object might be very small

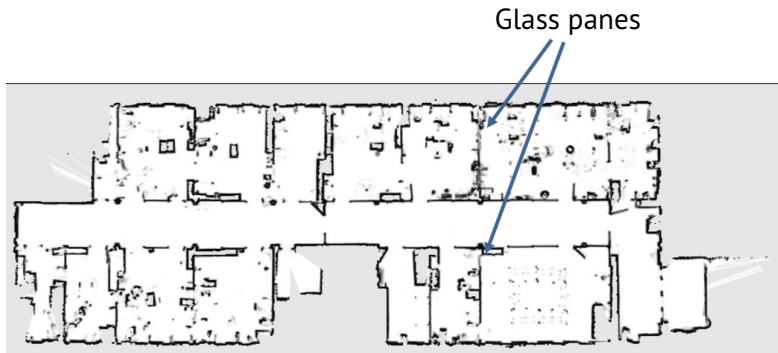
Example

Occupancy Map



Example

Reflection Map



Summary (1)

- Grid maps are a popular model for representing the environment
- Occupancy grid maps discretize the space into independent cells
- Each cell is a binary random variable estimating if the cell is occupied
- We estimate the state of every cell using a binary Bayes filter
- This leads to an efficient algorithm for mapping with known poses
- The log odds model is fast to compute

Summary (2)

- Reflection probability maps are an alternative representation
- The key idea of the sensor model is to calculate for every cell the probability that it reflects a sensor beam
- Given the this sensor model, counting the number of times how often a measurement intercepts or ends in a cell yields the maximum likelihood model
- This approach has a consistent sensor model for mapping and localization